Dataset Source and Study Scope

WasteSort | Goal Diggers | BCS34
Research progress submission - November 7, 2026

Study design

This submission uses secondary data for supervised waste-material image classification. The researchers obtained, audited and preprocessed an existing labeled dataset, trained models and evaluated their predictions. The group does not claim to have photographed the source items.

Dataset and attribution

The dataset is TrashNet, created by Gary Thung and Mindy Yang. The source archive contains 2,527 resized images in six classes: cardboard, glass, metal, paper, plastic and trash. The creators photographed individual items on white posterboard under sunlight and/or room lighting. Source: github.com/garythung/trashnet. The pinned archive, attribution and retrieval checksum are included in 01_Datasets.

Researcher preparation

The group retained 2,521 images after excluding six conflicting-label exact duplicates. Candidate duplicate groups remain in one partition. The recorded split contains 1,774 training, 371 validation and 376 test images. The submission includes CSV inventories, labels, split assignments and 1,764 HOG feature values per retained image. These extracted features are derivatives of secondary data; they are not newly collected primary observations.

Primary data scope

No primary dataset is used in this initial study. Campus-photo collection is a possible later evaluation phase. Results here describe the recorded TrashNet benchmark and do not establish campus performance.

Files and actual results

The source-code folder includes runnable Python code, trained models and executed notebook outputs. Initial_Model_Training_Results.pdf reports the original measured results, including 84.57% test accuracy and 81.22% macro F1 for MobileNetV2. Brief_Technical_Report.pdf explains the methodology, limitations and recommendations. Read COLAB_VERIFICATION.md before comparing the separate cloud run.

Source reference

Thung, G., and Yang, M. TrashNet dataset and project repository. https://github.com/garythung/trashnet . Pinned dataset archive: https://huggingface.co/datasets/garythung/trashnet/resolve/94cd17e/dataset-resized.zip . Retrieved October 20, 2026; source documentation checked November 7, 2026.